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REGULAR ARTICLE

# Interaction of phytohormone-producing rhizobia with sugarcane mini-setts and their effect on plant development

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**Abstract**
*Aims* Rhizobia are common members of plant microbiomes. This study aimed to evaluate if rhizobia can colonize sugarcane mini-setts and if and how they affect sprouting and morphological traits of plant development.
*Methods* *Bradyrhizobium sacchari* strain BR 10280$^T$ and *Rhizobium* sp. strain BR 10268 were inoculated on sugarcane mini-setts and sprouting and plant development were evaluated. Bacterial production of hormones and their effect on plant growth were studied using HPLC and GC-MS / SIM and sugarcane sprouting assays. Colonization was investigated by colony counting and CLSM. Sequence analysis of *recA*, *gyrB* and *rpoB* genes was applied to refine the phylogenetic classification of strain BR 10268.
*Results* BR 10268 had a positive effect on sugarcane mini-sett sprouting and shoot and root growth at 30 and 75 days after transplantation (DAT) and colonized sprouting mini-sett tissues. Few significant effects were observed for *B. sacchari* BR 10280$^T$. Cell-free BR 10268 supernatant was found to contain gibberellins (GAs) and low concentrations of indole-3-acetic acid (IAA) and trans-zeatin; treatment of mini-setts with $GA_3$ (0.15 or 1.5 μg $mL^{-1}$) affected sprouting in a similar manner as BR 10268 inoculation. BR 10268 was found to be phylogenetically close to *R. freirei*.
*Conclusions* *Rhizobium* sp. BR 10268 positively affects sprouting and growth of sugarcane, probably via production of plant hormonal substances, notably gibberellins, thus naturally occurring rhizobia may have potential as sugarcane inoculants.

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**Keywords** Endophytic bacteria · Plant growth promoting bacteria · Sprouting · Confocal laser scanning microscopy · Gibberellins · HPLC · GC-MS

**Abbreviations**

| | |
|---|---|
| CFU | Colony forming units |
| TY | Tryptone yeast extract |
| YMA | Yeast extract mannitol agar |
| DAI | Days after inoculation |
| DAT | Days after transplanting |
| HPLC | High performance liquid chromatography |
| GC-MS / SIM | Gas chromatography-mass spectrometry in selective ionic monitoring mode |
| O.D. | Optical density |
| $GA_x$/GAs | Gibberellin x / gibberellins |
| CLSM | Confocal laser scanning microscopy |

## Introduction

Sugarcane is an economically important crop used for the production of sugar and ethanol, which can be used as a renewable energy source in substitution of fossil fuel. Brazil is a major sugarcane producer for the domestic and foreign markets with a production of 657.18 million Mg in the season 2016–2017 (Conab 2018).

Sugarcane is planted through vegetative propagation and not by seeding. Traditionally, pieces of sugarcane stems containing approximately three nodes are used as planting material. Novel methods have been developed to reduce the use of large amounts of millable stems by using single nodes (mini-setts) to produce pre-sprouted seedlings. These methods possess several advantages, such as the reduction in 80% of the sugarcane stems used for renewing sugarcane fields (more than 20 Mg $ha^{-1}$ of stems may be used in traditional planting), enhanced planting homogeneity, improved disease control by the utilization of heat and fungicide treated seedlings (Landell et al. 2012). In this system, plantlets are produced in a nursery in a more efficient way, and then transplanted to the field.

The sprouting phase of mini-setts represents an opportunity to improve the performance of this material such as by the introduction of plant growth promoting bacteria. Examples of well-studied sugarcane growth promoting bacteria, which exert their effect via biological nitrogen fixation (BNF) and phytohormone production, are *Gluconacetobacter diazotrophicus*, *Herbaspirillum seropedicae*, *Herbaspirillum rubrisubalbicans*, *Nitrospirillum amazonense* and *Paraburkholderia tropica* (Oliveira et al. 2002, 2006; Schultz et al. 2017). Recently, a mix of these bacteria was shown to positively influence sugarcane mini-sett sprouting and initial root development (Santos et al. 2018).

In recent years, culture independent methods have been applied to sugarcane to uncover the diversity of microorganisms associated to the crop and to get an idea of its function by several research groups. Among others, Thaweenut et al. (2011), Fischer et al. (2012) and Burbano et al. (2011) demonstrated the presence of nitrogenase mRNA (*nifH*) related to rhizobia (genera *Rhizobium* and *Bradyrhizobium*) in tissues of field-grown sugarcane, which suggested a role of these bacteria in sugarcane-associated BNF. Later, a diverse collection of *Bradyrhizobium* and *Rhizobium* isolates was obtained from surface-disinfected sugarcane roots in Brazil (Rouws et al. 2014; Rouws et al. 2015). Genetic characterization of this collection also led to the proposal of a novel sugarcane-associated bacterium, named *Bradyrhizobium sacchari* (Matos et al. 2017). The widespread occurrence of sugarcane-associated rhizobia has since then been confirmed in several culture-independent studies (Souza et al. 2016; Dong et al. 2018; Yeoh et al. 2016). Rhizobia have also been isolated from maize, a C4 photosynthetic metabolism plant species like sugarcane (Gutiérrez-Zamora and Martınez-Romero 2001).

Natural occurrence and relevance of rhizobia in non-legumes have been extensively studied for the interaction between the rice (a C3 photosynthetic metabolism crop) and *Rhizobium leguminosarum* bv. *trifolii* in the Nile delta in Egypt. For that interaction, beneficial effects of inoculation on plant growth have been shown both in laboratory and field studies (for example Yanni et al. 1997; Yanni et al. 2001). Inoculation with *Bradyrhizobium* strains isolated in West Africa and Southeast Asia promoted growth of rice under controlled conditions (Chaintreuil et al. 2000; Piromyou et al. 2015). In rice, several rhizobia have been shown to invade root tissues from where they can subsequently reach the shoot via vascular tissue (Chi et al. 2005). Similarly, endophytic bacteria have been shown to be able to spread systemically in sugarcane via xylem vessels as well (James et al. 2001). However, a recent study showed that rhizobia inoculated on the root system can affect gene expression in rice shortly after inoculation and before shoot colonization, showing that

the plant can respond systemically to the presence of rhizobia without or before endophytic colonization (Wu et al. 2018). Another recent study involving genome analyses of 1314 bacterial strains suggested that commensal and beneficial relationships of Rhizobiales with plants, including non-legumes, represent an ancestral predisposition of this order and that specific genetic determinants for mutualism with legumes (for example *nod* genes) were acquired later in evolution (Garrido - Oter et al. 2018).

The present study aimed to evaluate the potential of two strains of rhizobia previously isolated from sugarcane to promote the sprouting and initial plant development of sugarcane propagated from mini-setts. The first is strain BR 10280$^T$, type-strain of the recently proposed novel species *B. sacchari*; the other is *Rhizobium* sp. BR 10268 (synonym P5–2), which was tentatively classified as *R. miluonense* based on 16S rRNA sequence analysis (Rouws et al. 2014). Growth promotion effects and structural and functional aspects of these rhizobia during their interaction with sugarcane are presented.

## Materials and methods

### Sprouting of inoculated mini-setts under greenhouse conditions

The experiment was initiated in May 2016. Sugarcane mini-setts were prepared from 1-year old sugarcane plants, cultivar RB 867515, grown at Embrapa Agrobiologia experimental station. Cultivar RB 867515 is one of the most used sugarcane cultivars in Brazil and is well adapted to low fertility soils, responding well to bacterial inoculation (Santos et al. 2018). Therefore, this cultivar was used in all experiments in this study. Before inoculation, mini-setts (approximately 2 cm long) were treated with fungicide (methyl N-(2-{[1-(4-chlorophenyl)-1H-pyrazole-3-yl] oxymethyl} phenyl)N-methoxy carbamate (Pyraclostrobin 250 g L$^{-1}$)) diluted in tap water at 0.1% for 3 min and then subjected to heat treatment (52 °C for 30 min in a water bath) to decrease the incidence of pathogenic fungi and naturally associated bacteria.

Strains BR 10280$^T$ and BR 10268 were grown at 28 °C in TY liquid culture medium (Vincent 1970) under shaking at 150 rpm. As a control, the five-strain sugarcane inoculant of Embrapa Agrobiologia was used (Reis et al. 2009). The respective strains (*Gluconacetobacter diazotrophicus* strain BR 11281$^T$, *Herbaspirillum seropedicae* strain BR 11335, *H. rubrisubalbicans* strain BR 11504, *Paraburkholderia tropica* strain BR 11366$^T$ and *Nitrospirillum amazonense* strain BR 11145) were grown in Dygs culture medium for 2 d at 30 °C (Baldani et al. 2014). All bacterial strains were cultivated to an optical density of O.D.$_{(600\ nm)}$ = 0.6–0.8, equivalent to 1 × 10$^9$ CFU mL$^{-1}$. After growth, each bacterial strain culture was individually diluted 100 times in autoclaved tap water and mini-setts were inoculated by immersing them in the resultant suspensions (containing approximately 1 × 10$^7$ CFU mL$^{-1}$ for each individual strain) for 1 h before planting. In the case of the five-strain inoculant, the 5 strains were 100 times diluted and mixed in the same volume of water for inoculation. The non-inoculated control consisted of mini-setts immersed in diluted 100 times diluted TY medium without bacteria.

After inoculation, mini-setts were planted in trays containing 13 kg of a sterilized mixture of washed sand and vermiculite (2:1 *v*/v), pH = 6.7, with no detectable nitrogen. For each treatment, four trays containing 15 mini-setts each were organized in the greenhouse in randomized blocks design and watered with sterilized water when necessary.

Emergence of shoots was accompanied daily during 23 days after inoculation (DAI). After this period, representative emerged plants of each treatment were selected and carefully transplanted to pots for prolonged growth in the greenhouse.

### Prolonged growth of sugarcane plantlets for 30 and 75 DAT under greenhouse conditions

Plantlets obtained from sprouted mini-setts in the greenhouse were planted in pots containing 3 or 10 L of the same sand/vermiculite mixture as described before. Plants in 3 L capacity pots (10 replicates per treatment) were collected 30 days after transplanting (DAT), and those transplanted into 10 L pots (10 replicates per treatment) were collected at 75 DAT. The substrate was supplemented with (g L substrate$^{-1}$): simple superphosphate (3.8), $K_2SO_4$ (0.23), agricultural limestone (0.9). Micronutrient solution (0.5 mL L substrate$^{-1}$) as described by Franco and Döbereiner (1967), modified by the addition of $MnSO_4$ (0.5 g L$^{-1}$) was also applied. For N fertilization, a solution of $(NH_4)_2SO_4$ (30 mg mL$^{-1}$) was prepared. All inoculation treatments and the low-N treatments received the equivalent of 40 kg N ha$^{-1}$ (60 mg N pot$^{-1}$ and 200 mg N pot$^{-1}$ for 3 L and 10 L pots respectively). The high-N treatments received the equivalent of 120 kg N ha$^{-1}$ (180 mg N pot$^{-1}$

and 600 mg N pot$^{-1}$ for 3 L and 10 L pots respectively). Therefore, the experiment consisted of five treatments: 1. High-N, non-inoculated; 2. Low-N, non-inoculated; 3. Low-N, five-strain inoculant; 4. Low-N, inoculated with *Rhizobium* sp. BR 10268; 5. Low-N, inoculated with *B. sacchari* BR 10280$^{T}$. Plants were organized in the greenhouse in completely randomized blocks.

### Testing the effect of *Rhizobium* sp. BR 10268 and $GA_3$ on sugarcane sprouting under laboratory conditions

For laboratory sprouting assays, strain BR 10268 was grown in TY broth as described above for greenhouse experiments. To obtain cell-free culture supernatant, cells were spun down in a centrifuge at room temperature and supernatant was sterilized using a syringe coupled to a 0.22 μm Millipore filter. The final inoculant formulations were prepared as follows: Bacterial cultures, culture supernatant or non-inoculated TY medium (used as negative control) were diluted 100 times in autoclaved tap water. In treatments to evaluate the effect of GAs on sprouting, required amounts of $GA_3$ (Caisson labs, USA) were added to the same formulation used for negative controls. After inoculation, mini-setts were planted in plastic recipients containing 500 g of the same sand/vermiculite mixture described before. Experiment 1 was performed with mini-setts prepared from stems from a sugarcane mill in São Paulo State in September 2016, using a completely randomized design with 8 replicates (96 experimental units) in a growth cabinet at 30 °C. Experiment 2 was carried out in March 2017 with stems from the Embrapa Agrobiologia experimental station in a randomized block design with 7 replicates. Experiment 2 was performed in plant growth room at a temperature of 30 °C. For both experiments, a 12 h photoperiod was applied with light provided by fluorescent tube lamps. Irrigation was performed every other day by adding 30 mL sterile tap water per recipient. A third experiment was performed in July 2019 to compare the effects of inoculation with BR 10268 and different concentrations of $GA_3$ (0.15, 1.5, 15 and 150 mg L$^{-1}$). This experiment was performed in a growth room at 30 °C with mini-setts from Embrapa Agrobiologia experimental station. This experiment was performed with 28 replicates in a completely randomized blocks design.

### Biometric evaluation of plant development

The following biometric variables were determined in this study: shoot length; total root length; shoot and root dry mass; stem base diameter; root volume; total leaf surface area; whole plant N accumulation and concentration ([N]).

Root volume was determined by the water displacement method; the root system of each plant after washing with running water to remove sand and vermiculite, was submerged in a graduated cylinder with water and the difference in volume before and after the submersion of roots was recorded (in mL).

For the dry matter determinations, tissues were dried at 65 °C in an oven with forced circulation until reaching constant mass and then weighed using a precision scale. After weighing, some samples were ground in a Wiley mill with 2 mm sieve for posterior N determinations. Shoot length was determined using a graduated ruler, measured from the base of the plant to the apex of leaf +1. The stem diameter was determined using a caliper.

Total leaf surface area was determined by scanning all green leaves of selected plants using a stationary leaf area meter (LI-3100C, LI-COR, Lincoln NE, USA). [N] determinations were performed using the Semi-micro Kjeldahl method as described (Urquiaga et al. 1992) at the Agricultural Chemistry Laboratory at Embrapa Agrobiologia. N accumulation was determined by multiplying dry mass and [N] values.

Root length data shown in Fig. 5 were obtained by scanning whole root systems of individual plants (28 plants treatment$^{-1}$) and analyzing the images using WinRhizo® software (Regent Instruments Inc., Canada).

### Evaluations of mini-sett bacterial colonization by counting and BOX-PCR

Bacterial colonization of mini-setts was evaluated under laboratory conditions performed with sugarcane stems obtained from the Embrapa Agrobiologia experimental station. Mini-setts were prepared and treated as described above. The experimental design was completely randomized, consisting of 2 treatments (inoculated and non-inoculated), 5 replicates and 2 collection dates at 7 and 20 DAI. The mini-setts were planted in plastic pots containing 500 g of a 2: 1 (v / v) autoclaved sand and vermiculite mixture placed in a growth cabinet at 30 °C with 12 h photoperiod.

At the collection dates, three mini-setts/treatment were carefully removed from the substrate and washed with sterile water to remove adhering substrate. The mini-setts were separated into three segments: sett, shoot and root. Samples were weighed using an analytical balance and then the shoot and root samples were immersed in 70% ethanol for 1 min, 2–3% sodium hypochlorite solution for 2 min, followed by 5 successive washes in sterile distilled water. To ensure the effectiveness of the disinfestation, aliquots from the last wash phase were plated and the disinfected shoots and roots were briefly placed on YMA medium plates which were incubated at 30 °C for 3 d to confirm effectiveness of the disinfection procedure. Shoot and root samples were then manually macerated in a mortar and diluted serially ($10^{-1}$ - $10^{-5}$) using sterile saline solution (NaCl 0.8%).

Mini-setts were immersed in pure ethanol and then flamed. Then the setts were peeled, removing all surface tissue using a flamed kitchen knife. The inner tissue thus obtained was weighed under sterile conditions, immersed again in pure ethanol, and flamed twice. These disinfected samples were homogenized in a blender previously disinfected with 2–3% sodium hypochlorite solution. As a sterility test, before the first use and between the different samples, the blender was operated after addition of 200 mL of sterile water and aliquots (50 μL) of this water were plated to verify the absence of bacteria. The mini-sett samples were ground in a volume of 200 mL of sterile water for approximately 3 min and the original samples were diluted down to $10^{-5}$ in relation to the original sample. For all plant sample dilutions, 50 μL aliquots were plated on TY plates, which were incubated at 28 °C for 3 d, when colonies could be quantified and visually compared to the inoculated strain.

To confirm the visual identification of re-isolated bacteria, some of the identified colonies, and also some colonies visually identified to be different, were compared to the inoculated strain BR 10268 at the genetic level. Colonies were picked with pipet tips and suspended in 100 μL ultrapure water and submitted to BOX-PCR, using the GoTaq DNA polymerase kit (Promega) and primer Box A1R (5'-CTACGGCA AGGCGACGCTGACG-3 ') (Versalovic et al. 1994). The PCR program was as follows: initial denaturation (95 °C for 10 min) followed by 35 amplification cycles (94 °C for 1 min, 53 °C for 1 min and 65 °C for 8 min) and final extention at 65 °C for 16 min. PCR products were separated on 2% agarose / TAE gels, which were stained with ethidium bromide. Gels were analyzed using BioNumerics 7.6 software (Applied Maths, Belgium), and clustering of profiles was performed using the Dice similarity coefficient and the UPGMA clustering method.

## Confocal laser scanning microscopy (CLSM)

For microscopy, a red-fluorescent derivative of BR 10268 was constructed by transformation with plasmid pLMB426, using electroporation. Plasmid pLMB426 was gently provided by Dr. Philip Poole and carries the mCherry gene downstream of the constitutive Ptac promotor and a gentamycin gene as a selectable marker (Rott 2012). The transformed strain, designated as BR 10268(mChe), was cultivated using liquid or solid TY medium amended with gentamycin (40 μg $mL^{-1}$). Mini-setts were inoculated with a suspension containing $10^7$ CFU BR 10268(mChe) cells or left non-inoculated, using procedures described above, and incubated in pots with sterilized sand humidified with autoclaved water. For microscopic analysis, mini-setts were removed from the substrate and rinsed with water to remove sand particles. Newly emerged sett roots were excised and investigated without sectioning. To investigate internal colonization, mini-setts we rinsed with pure ethanol and flamed to kill surface-associated bacteria. Slices of internal tissues were obtained by sectioning setts with a flamed kitchen knife, and then mounted on microscope slides. Tissues were investigated under an LSM 700 confocal laser scanning microscope (Carl Zeiss, Germany) at 3, 5 and 8 DAI. Red and green fluorescence were acquired using 555 nm (channel 1) and 488 nm (channel 2) laser lines respectively. Transmitted light from channel 1 was also acquired. Laser intensities (2%) and gain settings for both channels were set at equal values to enable comparisons of signals and prevent bias.

## Cultivation of BR 10268 in defined liquid medium for phytohormone determinations

Strain BR 10268 was cultivated in modified RDM (*Rhizobium* defined medium; Bishop et al. 1976) containing (g $L^{-1}$): D-mannitol (50); $KH_2PO_4$ (1.00); $K_2HPO_4$ (1.00); $NH_4CL$ (0.11); $MgSO_4.7H_2O$ (0.25); NaCL (0.25); $CaCL_2.2H_2O$ (0.025); sodium glutamate (0.11); $Fe^{2+}$EDTA (0.0328); $Na_2MoO_4.2H_2O$ (0.0004); $MnSO_4.4H_2O$ (0.00047); $H_3B0_3$ (0.00056); $CuSO_4.5H_2O$ (0.000016); $ZnSO_4.5H_2O$ (0.000048); biotin (0.00001); pyridoxin-HCl (0.00002). A single

colony was inoculated in 50 mL of modified RDM medium in a 250 mL capacity Erlenmeyer flask, and grown on a rotary shaker at 150 rpm and 30 °C. When the culture attained a density of $3 \times 10^9$ CFU $mL^{-1}$, 25 mL of bacterial suspension was centrifuged for 10 min, 4000 rpm at 4 °C to sediment the cells. Supernatant was sterilized by filtration through a 0.22 μm Millipore filter. Aliquots (1.5 mL) of the supernatant were vacuum dried in a SpeedVac apparatus (Labconco) and used for phytohormone analyses.

### Phytohormone analyses

Phytohormone analyses were performed by high performance liquid chromatography (HPLC) followed by gas chromatography-mass spectrometry in selective ionic monitoring mode (GC-MS / SIM). Dried BR 10268 supernatant samples were dissolved in 1.5 mL methanol and subjected to reverse phase HPLC (C18-HPLC) using a 3.9 mm × 300 mm Bondapak C18 column (Waters Corp., USA) and eluted at 1.5 mL $min^{-1}$ with the following mobile phases: 0 to 5 min, 28% methanol in 1% aqueous acetic acid (isocratic); 5 to 35 min, linear gradient from 28% to 86% methanol; 35 to 36 min, 86% to 100% methanol; 36 to 40 min, 100% methanol (isocratic). A total of 48 fractions of 1.5 mL each were collected. The fractions were prepared for injection in a gas chromatograph / mass spectrometer (GC / MS) with selective ion monitoring mode (SIM) (GC system 6890 N and selective network mass detector 5973; Agilent Technologies, USA). For each type of substance found, 1 μL of sample was injected into a DB-1 capillary column with 30 mm × 0.25 mm ID and 0.25 μm film thickness (J & W Scientific Co., USA). The temperature of the GC oven was programmed for 1 min at 60 °C, then increasing at 15 °C $min^{-1}$ to attain 200 °C, followed by a 5 °C $min^{-1}$ increase to a final temperature of 285 °C. The carrier gas (helium) was maintained at a pressure of 30 kPa. The GC was directly connected to a selective mass detector with an interface and source temperature of 280 °C, an ionization voltage of 70 V and a residence time of 100 ms. The complete scan mode (the first test) and the three main ions of the internal standards trans-zeatin, IAA and GAs, supplemented with the detected compounds, were monitored simultaneously. Retention time was determined using hydrocarbon standards to calculate the KRI (Kovat Retention Index) value, and the quantification of the substances was based on HPLC peak ratios of each non-deuterated (extracted) compound to the deuterated standards.

### Statistical procedures

Analyses of variances were performed after testing the assumptions of normality and homogeneity of residue variances using Shapiro-Wilk and Bartlett tests. The means of treatments were compared using the LSD or the Scott-Knott test. The software R (R Core Team 2018) was used in all the statistical analyses at 5 or 10% probability).

### Phylogenetic analyses

Housekeeping genes were amplified by polymerase chain reaction (PCR) following instructions as described previously by Vinuesa et al. (2005) (*recA*) and Martens et al. (2008) (*gyrB* and *rpoB*). PCR products were sequenced in two directions using the same primers used for PCR amplification. For *nifH*, PCR was performed using primers nifHF and nifHI (Laguerre et al. 2001). The PCR fragment (approximately 800 bp) obtained with the *nifH* primers was cloned using the pGEM-T easy vector system using T4 ligase (Promega). The ligation product was transferred to *Escherichia coli* DH10B electrocompetent cells by electroporation. After blue-white screening of transformant colonies, plasmid DNA was prepared and submitted to PCR with primers pGEM-F3000 and pGEM-R208 in order to amplify the plasmid insert (de Novais et al. 2014). Amplicons were sequenced with the same primers. After post reaction clean-up, PCR products were submitted to sequencing using the Big Dye™ Terminator v 3.1 cycle sequencing kit (Applied Biosystems) and an automatic DNA sequencing manifold ABI 3730xl (Applied Biosystems). Sequences were assembled and checked for quality using BioNumerics 7.6 and deposited in the genbank under accession numbers given in Fig. 6 and Fig. S4.

Sequence alignments with sequences from type-strains from the genus *Rhizobium* were generated using the Muscle tool in Mega 6 software (Tamura et al. 2013). For phylogenetic reconstruction, Mega 6 was used with the 'complete deletion' option, using the maximum-likelihood (ML) construction method (Felsenstein 1985). The most suitable model for the calculation of phylogenetic trees was determined for sequence alignment using the integrated model selection tool of Mega 6. The reliability of the

phylogenetic tree topology was evaluated using the bootstrap method with 500 pseudoreplicates.

## Results

### Effect of inoculation with rhizobia on mini-sett sprouting

Sprouting of sugarcane mini-setts in the greenhouse was evaluated by daily counting of newly emerged shoots. Improved sprouting was observed for setts inoculated with *Rhizobium* sp. BR 10268, with first sprouts being observed at 9 DAI and superior sprouting persisted throughout the experiment. Slowest sprouting was observed for *B. sacchari* BR 10280$^T$ and non-inoculated and five-strain inoculated treatments showed intermediate values (Fig. 1). Representative sprouted mini-setts of all treatments were transplanted to pots.

In the greenhouse sprouting experiment, the exact moment of shoot and root emission below the substrate surface could not be accompanied. Therefore, the effect of inoculation with BR 10268 on mini-sett sprouting was studied in two laboratory experiments at 3, 6, 8, 10 and 14 DAI (Fig. S1), where sprouting mini-setts were uprooted and evaluated in more detail. In both experiments, average shoot length and mass tended to increase after inoculation. Increases of 30% and 104%, respectively for shoot length and dry mass, were observed at 8 and 10 DAI (experiment 1). Increases of shoot length (136%) and dry mass (131%) were also found on 3 and 6 DAI, respectively for experiment 2. The lack of statistical difference at several collection points were related to high levels of variability related to natural heterogeneity among sugarcane mini-setts obtained from field-grown sugarcane stems. Significant effects of inoculation on root variables were not observed in any occasion and average root dry mass values varied between 17% decrease (Experiment 2 at 3 DAI) and 14% increase (Experiment 1 at 14 DAI) (Fig. S1).

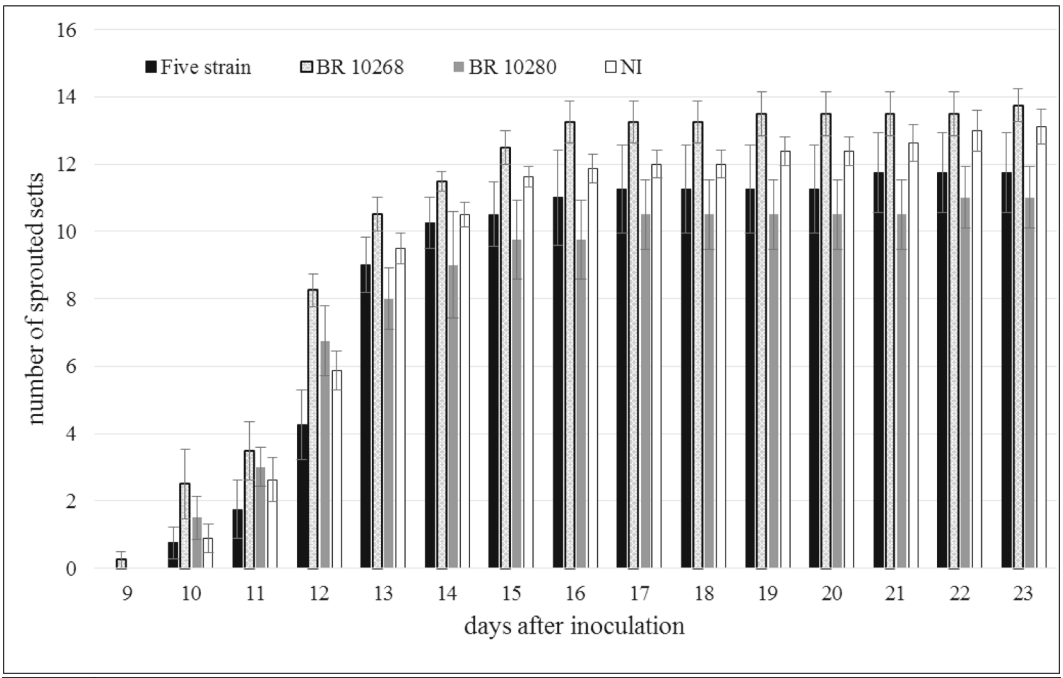
### Growth variables of inoculated sugarcane in the greenhouse at 30 and 75 DAT

At 30 days after transplanting sprouted mini-setts to pots, the positive effect of inoculation with strain *Rhizobium* sp. BR 10268 on shoot growth persisted and shoot length and dry mass were superior to the low-N control (Fig. 2a and b). The high-N treatment displayed somewhat reduced root development in comparison to the low-N and inoculation treatments, as determined by root volume and dry mass evaluations (Fig. 2c and d).

At 75 DAT, different from 30 DAT, the high-N treatment as well as inoculation with the five-strain inoculant and BR 10268 presented higher root volumes than the low-N treatment. BR 10268 also had a positive effect on root mass (Fig. 3a and b). With respect to shoot growth variables, stem diameter was positively affected in the high-N control, but mainly by inoculation with BR 10268 and the five-strain inoculant, when compared to the low-N control (Fig. 3c). Also total leaf surface area was increased in the high-N control and after inoculation with BR 10268 and the five-strain inoculant, in comparison to the low-N control (Fig. 3d). Shoot dry mass was highest for the high-N control and lowest for the low-N treatment and inoculation with BR 10268 and the five-strain inoculant led to intermediate values (Fig. 3e). No clear effects were observed after inoculation with BR 10280$^T$. Apart from the high-N treatment, which was superior, the total N accumulated by whole plants was similar among inoculated treatments and the low-N control (Fig. 3f). However, the N concentration of plant tissues was unaffected or even reduced (for strain BR 10268) in inoculated plants, when compared to the low-N control (Fig. 3g).

### Colonization of mini-setts by strain BR 10268

The colonization by strain BR 10268 of inoculated sprouting mini-setts, obtained from field-grown sugarcane stems and grown in sterilized substrate, was investigated by dilution plating of tissue extracts. After plating aliquots of extracts of surface disinfected sett tissues at 7 and 20 DAI, bacterial colonies could be observed for both BR 10268-inoculated and non-inoculated treatments. This was expected, since the mini-setts used were obtained from field-grown sugarcane plants and thus colonized by native bacteria. Nevertheless, visual inspection of plates permitted differentiating BR 10268 colonies from native bacteria. The former were present only in the inoculated treatment, but were absent in the non-inoculated treatment, in which only native bacteria were present (Table 1). Thirty representative colonies visually identified as BR 10268, along with some native isolates obtained from mini-setts and roots of both inoculated and non-inoculated plants, were submitted to BOX-PCR genetic analyses and compared to the profile




| Treatment/ DAI | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Five strain | a | b | a | c | ab | a | ab | ab | ab | ab | ab | ab | ab | ab | ab |
| BR 10268 | a | a | a | a | a | a | a | a | a | a | a | a | a | a | a |
| BR 10280$^{T}$ | a | ab | a | ab | b | a | b | b | b | b | b | b | b | b | b |
| NI | a | b | a | bc | ab | a | ab | ab | ab | ab | ab | ab | ab | ab | ab |

**Fig. 1** Sprouting of inoculated and non-inoculated sugarcane mini-setts under greenhouse conditions. Mini-setts of sugarcane cultivar RB 867515 were inoculated (Five strain, BR 10268 or BR 10280$^{T}$) or left non-inoculated (NI) and planted in trays containing autoclaved substrate. Each treatment consisted of four trays (replicates) containing 15 mini-setts each; the non-inoculated treatment consisted of 8 trays of 15 mini-setts. Sprouting was scored daily and average values are shown in the form of a bar graph. Standard error is represented by error bars. Different letters shown in the table below the graph indicate significant difference (LSD, $p < 0.1$) among treatment averages, for each time point

of BR 10268 genomic DNA. This analysis confirmed that the visually identified isolates were genetically identical to BR 10268, whereas bacteria identified as native were not (Fig. S2), thus confirming the visual identification. Comparing the two collection times at 7 and 20 DAI, bacterial densities dropped $10^2$-fold in mini-setts and around 10 times for roots at 20 DAI when compared to 7 DAI (Table 1).

CLSM analyses showed abundant red fluorescent bacteria colonizing the surface of newly formed roots on inoculated mini-setts at 5 DAI, where they seemed to be concentrated at junctions between epidermal cells and also on root hairs (Fig. 4a and b). Endophytic colonization could be shown in stem (mini-sett) tissue, more specifically in intercellular spaces between parenchyma cells and in protoxylem vessels of flamed and longitudinally sliced mini-setts at 3 and 8 DAI (Fig. 4c–f). No fluorescent bacteria were ever observed in non-inoculated control plants.

## BR 10268 produces several phytohormones in defined liquid medium

Chemical analyses were performed to discover if strain BR 10268 is able to synthesize plant growth regulating substances when growing in a defined culture medium. HPLC followed by GC-MS / SIM analyses showed the presence of several substances, mainly GAs $GA_1$, $GA_3$, $GA_4$, $GA_7$, $GA_9$, $GA_{12}$ and $GA_{20}$ (total of 11 ng $mL^{-1}$), IAA (3 ng $mL^{-1}$) and small amounts of the cytokinin trans-zeatin (0.3 ng $mL^{-1}$) (Table 2 and Fig. S3), which may explain the growth promotion effect of strain BR 10268.

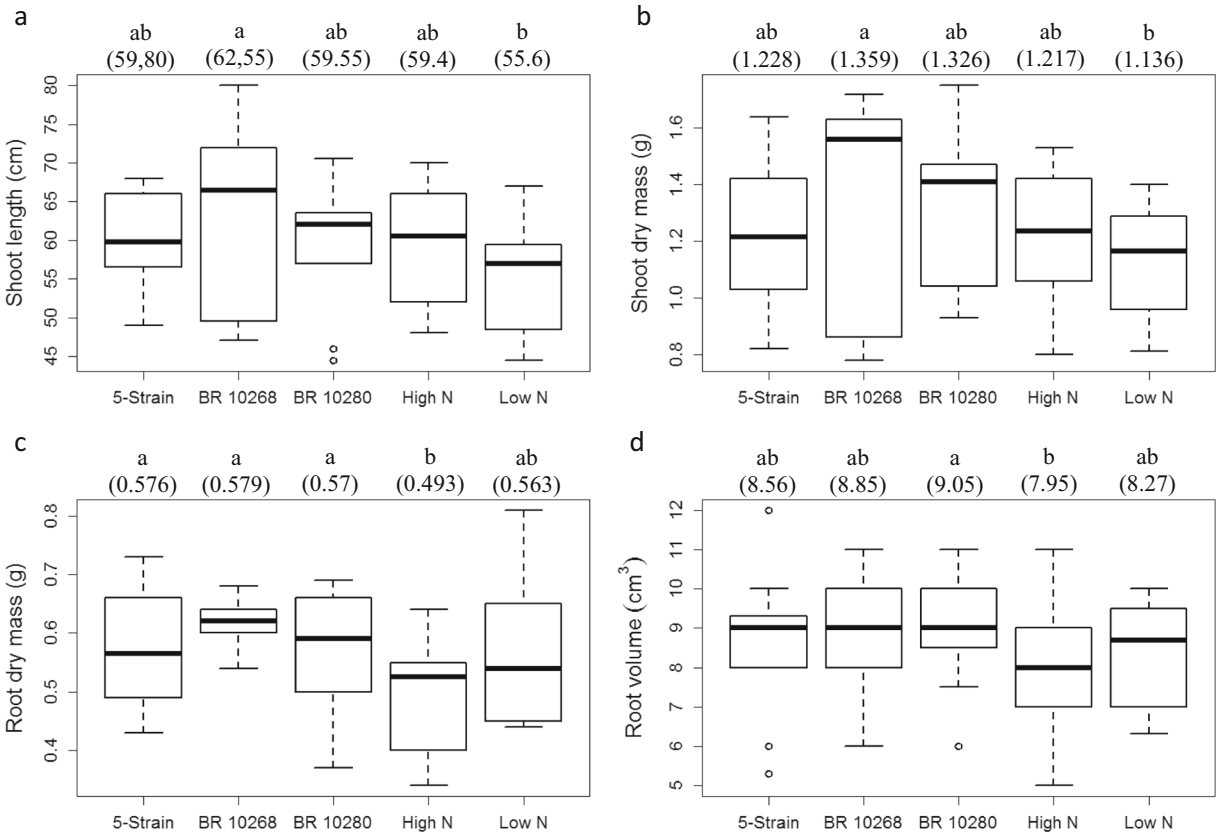



**Fig. 2** Effect of the inoculation of five-strain bacterial inoculant, *Rhizobium* sp. BR 10268, *B. sacchari* BR 10280$^T$ or N application (low-$N = 60$ mg N pot$^{-1}$; high-$N = 180$ mg N pot$^{-1}$) on plant growth variables at 30 DAT, under greenhouse conditions. Inoculated treatments received the low-N dosage. Average values of 10 replicates are shown, each consisting of a single plant in a 10 L pot, are shown in box-plot graph format. Median values are shown as horizontal lines inside boxes. Numbers shown above graphs represent average values and different letters indicate significant difference (LSD, $p < 0.1$)

## Similar effect of $GA_3$ and BR 10268 on sugarcane sprouting

Since GAs are well known to promote shoot growth in several plant species, we compared the effect of inoculating mini-setts with BR 10268 to the effect of exposing them to different physiologically relevant $GA_3$ concentrations at the time of planting. At five and six days after planting, the highest concentrations of $GA_3$ (15 and 150 μg mL$^{-1}$) showed a promoting effect on shoot emergence which was similar to, or superior to inoculation with BR 10268 (Fig. 5a). However, from 7 DAI onwards low concentrations of $GA_3$ (0.15 and 1.5 μg mL$^{-1}$) had similar beneficial effects on shoot length and after 10 and 11 DAI, both highest $GA_3$ concentrations (15 and 150 μg mL$^{-1}$) performed worse, whereas BR 10268 and the lower concentrations of $GA_3$ continued having similar effects. This result was also confirmed by dry shoot mass determinations (Fig. 5b). On the other hand, root length and dry mass at 11 DAI were most benefitted by application of higher $GA_3$ concentrations (Fig. 5c). Inoculation with BR 10268 did not affect root growth significantly during this initial growth phase, confirming data presented in Fig. S1.

## Phylogeny of strain BR 10268

To improve the phylogenetic classification of strain BR 10268, the DNA sequences of its *recA*, *rpoB* and *gyrB* genes were compared to type-strains of phylogenetically related *Rhizobium* species (Fig. 6, Fig. S4). Analyses based on all three genes placed strain BR 10268 in a clade close to *Rhizobium miluonense* CCBAU 41251$^T$ (Gu et al. 2008), consistent with the previously

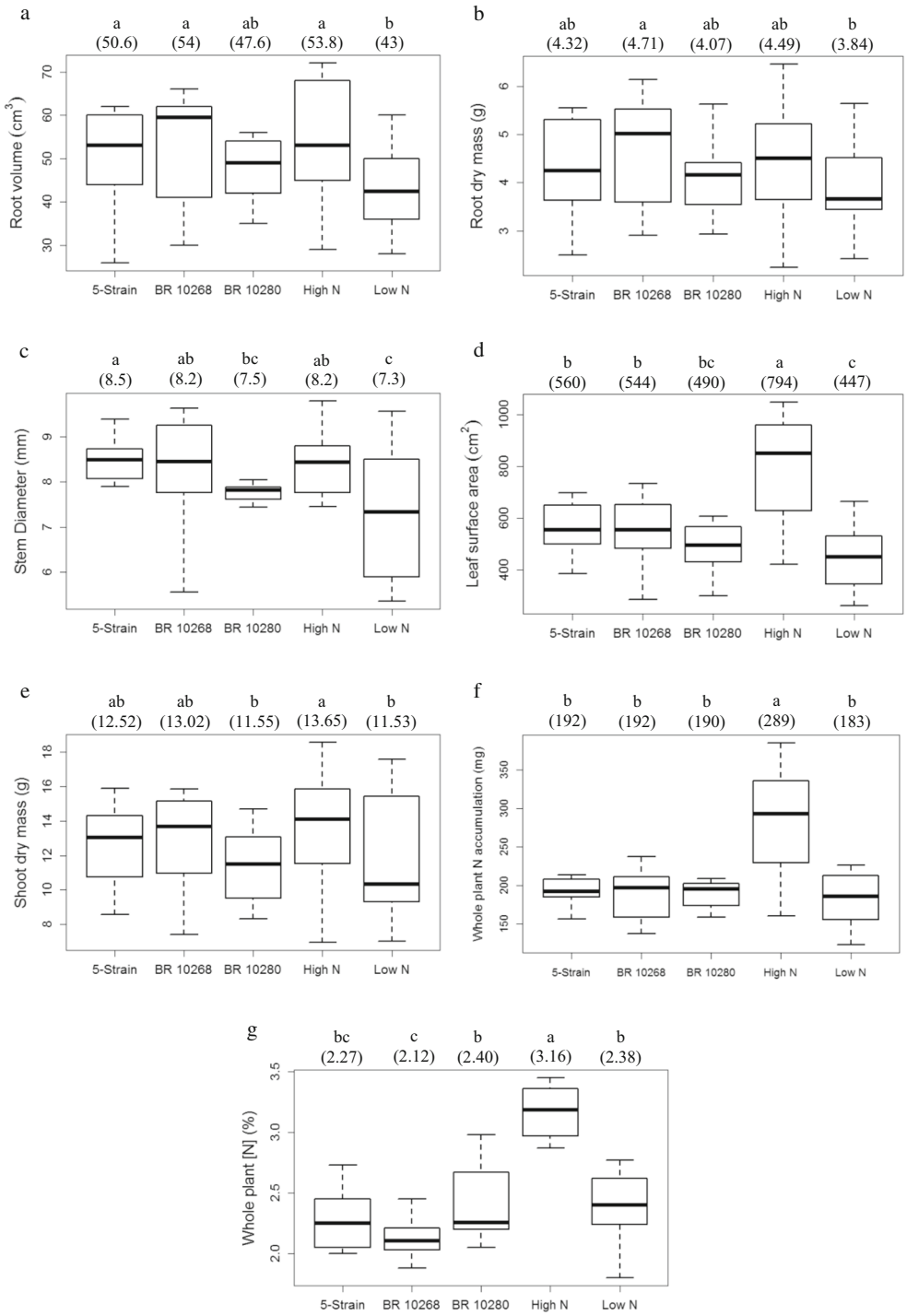



**Fig. 3** Effect of the inoculation of five-strain bacterial inoculant, *Rhizobium* sp. BR 10268, *B. sacchari* BR 10280$^{T}$ or N application (low-N = 180 mg N pot$^{-1}$; high-*N* = 600 mg N pot$^{-1}$) on plant growth variables of sugarcane plants at 75 DAT under greenhouse conditions. Inoculated treatments received the low-N dosage. Average values of 10 replicates are shown, each consisting of a single plant in a 3 L pot. Median values are shown as horizontal lines inside boxes. Numbers shown above graphs represent average values and different letters indicate significant difference (LSD, $p < 0.1$)

**Table 1** Densities of BR 10268 cells re-isolated from different surface-disinfected tissues of sprouting sugarcane mini-setts at seven and 20 DAI

| | 7 DAI | | | 20 DAI | | |
|---|---|---|---|---|---|---|
| | shoot | mini-sett | root | shoot | mini-sett | root |
| Inoculated | n.d. | $1.8 \pm 0.9 \times 10^{7}$ [a] | $4.9 \pm 2.7 \times 10^{5}$ | n.d. | $2{,}1 \pm 0{,}7 \times 10^{5}$ | $5{,}6 \pm 4{,}5 \times 10^{4}$ |
| Non-inoc. | n.d. | n.d. | n.d. | n.d. | n.d. | n.d. |

[a] Bacterial colonies were quantified by dilution plating and values are expressed as cfu $g^{-1}$ ± standard error of 3 replicates. n.d.: not detected

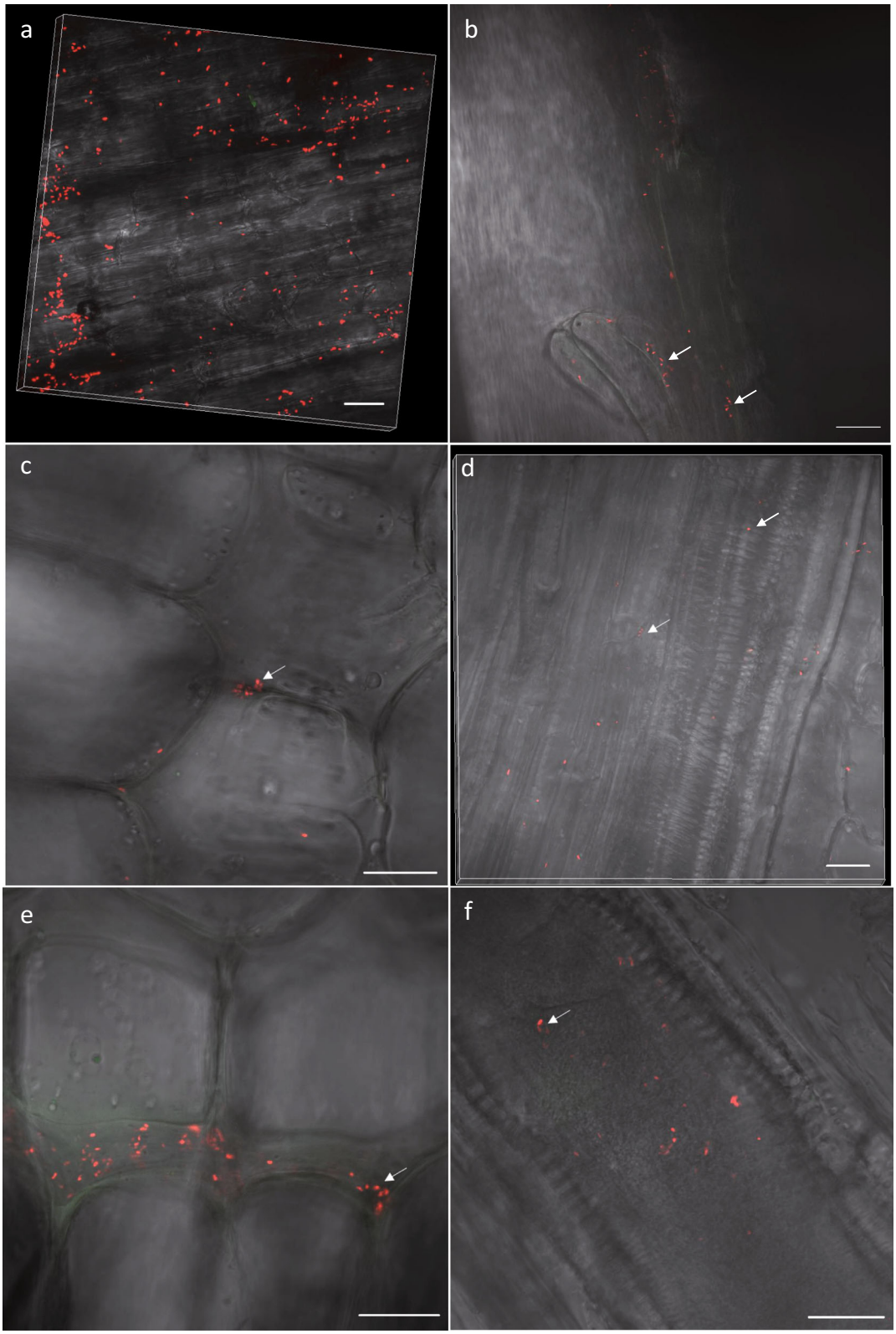
described phylogeny based on 16S rRNA (Rouws et al. 2014). However, the closest phylogenetic neighbor was *Rhizobium freirei* PRF 81 (Dall'agnol et al. 2013), both when using individual gene phylogenies (Fig. S4) and with concatenated sequences of these three genes (93.37% nucleotide sequence similarity) (Fig. 6). An amplicon of expected size (slightly smaller than 800 bp) could be obtained with *nifH* PCR primers, but sequencing of this fragment with the same primers was unsuccessful. After cloning the fragment, sequencing with primers flanking the insert was successful, but showed that the cloned fragment was not *nifH* (data not shown).

## Discussion

Very well known for their role as nitrogen fixing endosymbionts in legume root nodules, rhizobia are common soil bacteria that are frequently free-living saprophytes (Poole et al. 2018). Rhizobia are frequently reported as common rhizosphere and plant-associated bacteria, including non-legumes, and have been considered as part of the plant core microbiome (Rivilla et al. 2017). Non-legumes have been shown to benefit from BNF mediated by associative bacteria in crops like sweet potato, sugarcane, and rice (Yoneyama et al. 2017; Urquiaga et al. 1992, 2012; Bei et al. 2013). Nitrogenase (*nifH*) mRNA phylotypes related to alpha-rhizobia have been detected in tissues of field-grown sweet potato and sugarcane, showing in situ nitrogenase gene expression and thus suggesting that rhizobia may actively contribute to BNF in association to non-legumes (Terakado-Tonooka et al. 2008; Fischer et al. 2012; Thaweenut et al. 2011). Nevertheless, inoculation studies of non-legumes with strains of rhizobia or other associative bacteria, have indicated that the effects of inoculation on plant growth are not related to BNF and suggest that phytohormone production or other plant growth promoting effects are important (Biswas et al. 2000; Yanni et al. 2001; Schultz et al. 2017). Inoculation experiments with *Arabidospis thaliana* and diverse members of the order Rhizobiales recently indicated that the capability to promote plant growth is a common trait among these bacteria, even when they do not carry nodulation and nitrogen fixation genes (Garrido-Oter et al. 2018). Although rhizobia have been previously isolated from sugarcane, no studies have been performed to evaluate these bacteria with respect to sugarcane growth promotion.

In the present study, positive effects on shoot growth were observed during the first few days of sprouting after inoculation with *Rhizobium* sp. BR 10268, whereas there was no effect on root growth during the first 10 days after inoculation. Inoculation with *B. sacchari* BR 10280$^{T}$ did not have any clear positive effects on plant growth variables (Figs. 2 and 3). Nevertheless, *Bradyrhizobium*-related bacteria have been frequently reported as associates of non-legumes, including sugarcane and other grass-like species, suggesting an important role in the naturally growing plants (Videira et al. 2013; Fischer et al. 2012; Souza et al. 2016). Therefore it will be worthwhile to study the growth promotion potential of such bacteria in more detail. For example, growth promoting action might be investigated in more long-term field experiment.

During the sprouting phase of sugarcane setts, growth of the newly formed plant is supported by nutrient reserves from the sett tissue (Dillewijn 1952). Therefore, the initial shoot growth promotion effect caused by BR 10268 was probably not due to increases in available nutrients, for example by BNF, but rather to a growth modulation effect exerted by bacterial inoculation, such as bacterial production of phytohormones (Cassán et al. 2014; Garrido-Oter et al. 2018). The beneficial effect of strain BR 10268 on shoot growth, in comparison to the low-N treatment, was also observed at 30 DAT. Inoculation of sugarcane mini-setts with a five-strain


a
b
c
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e
f

◀ **Fig. 4** CLSM images of BR 10268(mChe) cells colonizing surface and internal tissues of sprouting sugarcane mini-setts. At 5 DAI, abundant red fluorescent bacterial cells were visualized associated to the root surface (**a**) and root hairs (**b**) 1 cm distant from the root tip. At 3 DAI, before sett root emission, bacteria could already be observed internally inside surface-disinfected setts, associated to parenchymatic (**c**) and vascular tissue (**d**). At 8 DAI, bacteria could be observed inside sett tissue at higher numbers, mainly in intercellular spaces between parenchyma cells (**e**) and also inside xylematic elements (**f**). Images are overlays of red (BR 10268(mChe) and green (plant cell wall autofluorescence) fluorescence and transmitted light signals. Single confocal images (**b**, **c**, **e** and **f**) or maximum intensity projections of three dimensional models contructed from a stack of confocal images (**a**, **d**) are shown. Arrows indicate observed bacteria. Scale bars represent 20 μm

inoculant was recently found also to improve sprouting, although this effect was not observed in the present study (Fig. 1 and Santos et al. 2018). In the study of Santos et al. (2018), the bacterial inoculant was applied as a peat formulation, whereas in the present study, freshly grown bacteria were suspended directly in sterile water, a difference which may have contributed to a different result. Substances like humic acids, which are present in peat, may have a synergistic effect on plant growth together with inoculated bacteria (Canellas et al. 2013). The five-strain sugarcane inoculant was formulated based on the results of inoculation experiments that showed that these five strains together performed better than each of them individually (Oliveira et al. 2006). This observation is now widely supported by research data and inoculation of mixed populations of microorganism with different functionalities to improve efficiency of crop production is a trend (Busby et al. 2017).

After more prolonged growth in the greenhouse, at 75 DAT, apart from shoot growth benefits observed at sprouting and at 30 DAT, there was a tendency of increased root development for the BR 10268-inoculated plants. At this point, root dry mass and volume were significantly increased in comparison to the low-N control and similar to the high-N treatment. At this point, both the five-strain inoculant and strain BR 10268 increased stem diameter and leaf surface area in comparison to low-N plants, but no significant effects could be found on shoot dry mass (Fig. 3).

**Table 2** Detection and quantification of phytohormones produced by BR 10268 grown in a defined culture medium by HPLC followed by GC-MS / SIM

| HPLC fraction | RT | Sample/standard | Substance | KRI | m/z (%, relative intensity of base peak) | | | concentration | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | (ng mL$^{-1}$) | (nmol L$^{-1}$) |
| 10 ~ 13 | 24.61 | Sample | $GA_1$ | 2650 | 507 (100) | 490 (7) | 449 (19) | 2.3 | 6.6 |
| | | Standard | $[^2H_2]GA_1$ | 2650 | 508 (100) | 493 (9) | 450 (21) | | |
| 11 ~ 15 | 24.59 | Sample | $GA_3$ | 2741 | 505 (100) | 489 (20) | 444 (8) | 3.3 | 9.5 |
| | | Standard | $[^2H_2]GA_3$ | 2741 | 506 (100) | 491 (10) | 445 (7) | | |
| 29 ~ 31 | 23.71 | Sample | $GA_4$ | 2515 | 285 (100) | 225 (66) | 263 (63) | 2.1 | 6.3 |
| | | Standard | $[^2H_2]GA_4$ | 2515 | 286 (100) | 227 (71) | 265 (68) | | |
| 35 ~ 38 | 24.50 | Sample | $GA_7$ | 2498 | 521 (100) | 488 (20) | 506 (8) | 0.6 | 1.8 |
| | | Standard | $[^2H_2]GA_7$ | 2498 | 524 (100) | 490 (20) | 507 (6) | | |
| 38 ~ 39 | 23.54 | Sample | $GA_9$ | 2315 | 298 (100) | 270 (70) | 228 (50) | 1.0 | 3.0 |
| | | Standard | $[^2H_2]GA_9$ | 2315 | 300 (100) | 272 (74) | 229 (50) | | |
| 40 ~ 44 | 22.01 | Sample | $GA_{12}$ | 2310 | 419 (100) | 404 (90) | 373 (68) | 0.3 | 0.9 |
| | | Standard | $[^2H_2]GA_{12}$ | 2310 | 420 (100) | 405 (92) | 375 (70) | | |
| 21 ~ 26 | 23.60 | Sample | $GA_{20}$ | 2501 | 419 (100) | 376 (47) | 404 (8) | 1.4 | 4.2 |
| | | Standard | $[^2H_2]GA_{20}$ | 2501 | 420 (100) | 377 (50) | 405 (10) | | |
| 15 ~ 19 | 7.70 | Sample | IAA | 2832 | 318 (100) | 302 (37) | 288 (12) | 3.0 | 17.1 |
| | | Standard | $[^{13}C_6]$IAA | 2832 | 319 (100) | 304 (35) | 290 (11) | | |
| 22 ~ 25 | 11.70 | Sample | Zeatin | 2832 | 229 (100) | 215 (22) | 185 (9) | 0.3 | 1.4 |
| | | Standard | $[^2H_2]$Zeatin | 2832 | 230 (100) | 216 (20) | 186 (7) | | |

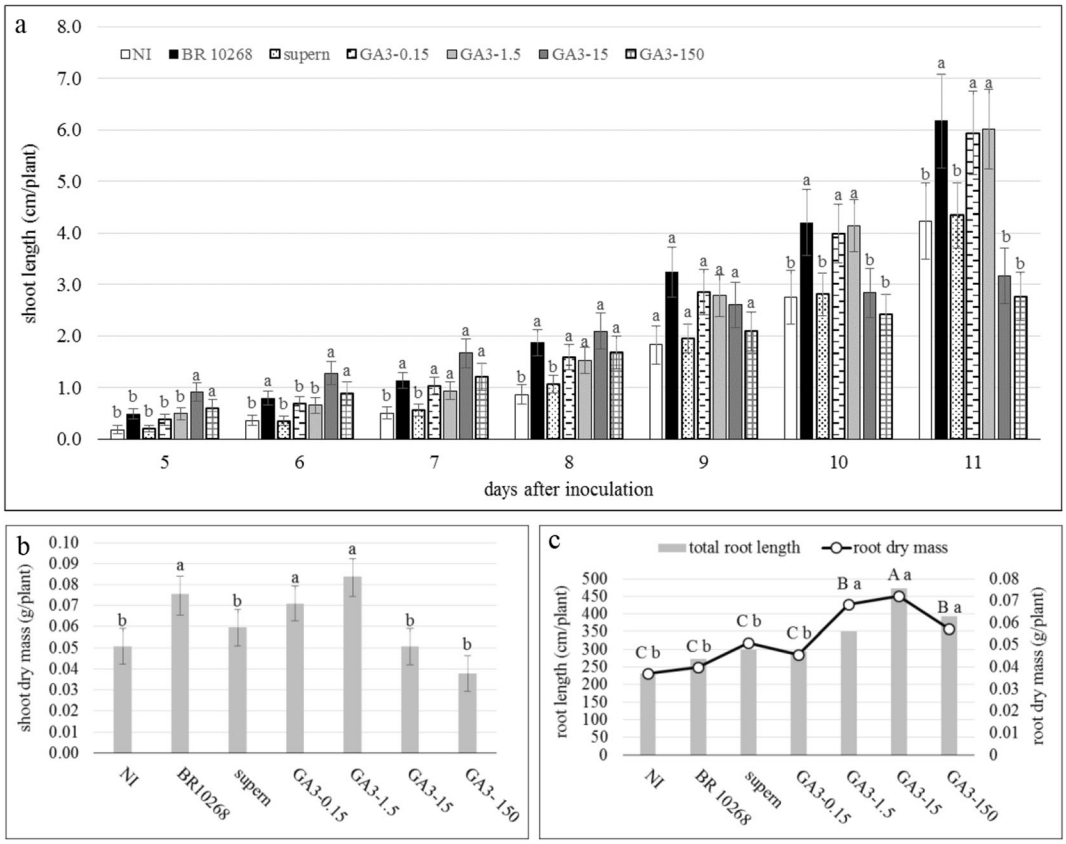


**Fig. 5** Comparison of the effect of inoculation with strain BR 10268 and treatment with increasing concentrations of gibberellin $GA_3$ on sugarcane sprouting. Heat and fungicide-treated sugarcane mini-setts (28 per treatment) were inoculated in watery solutions containing 100 times diluted BR 10268-containing broth (final concentration $1 \times 10^7$ CFU $mL^{-1}$), culture supernatant (supern), or fresh broth (NI). Alternatively, mini-setts were immersed in solutions containing 100 times diluted fresh culture broth, amended with different concentrations of $GA_3$. Treated setts were planted in pots and length of emerged sprouts was evaluated daily (**a**). At 11 days after planting, plants were harvested and the shoots were dried and their mass determined (**b**). Roots were separated and total root length was evaluated using an image scanner and WinRhizo® image-analysis software; roots were then dried before their mass was determined (**c**) Standard error levels are indicated by bars and letters above bars indicate statistical difference (Scott-Knott, $p < 0.05$). In (**c**), upper-case and lower-case letters refer to 'root length' and 'dry root mass' respectively; error bars were omitted from (**c**) for esthetic reasons

Although absolute N content of plant tissues was not influenced by treatments (with exception of the high-N treatment, which was superior), the N concentration was lower in all tissues of inoculated plants at 75 DAT. This supports that bacterial inoculation induced plant growth not via BNF, but via alternative mechanisms, possibly plant-hormone production, increasing biomass, but not directly contributing to N nutrition. In line with this, strain BR 10268 did not display nitrogenase activity in semi-solid N-free culture medium as determined by acetylene reduction assay (ARA).

A possible role of growth-modulating substances in plant growth stimulation was supported by the observation that cell-free culture supernatant of BR 10268 contained phytohormones, mainly GAs ($GA_1$, $GA_3$, $GA_4$, $GA_7$, $GA_9$, $GA_{12}$ and $GA_{20}$), some IAA and very low concentrations of the cytokinin trans-zeatin. GAs are generally reported to act at nanomolar concentrations in root development and micromolar concentrations during shoot development of different plant species (Tanimoto 2005; Hedden and Sponsel 2015).

We found high similarity in sprouting response to inoculation with BR 10268 and treating mini-setts with lower concentrations of $GA_3$ (0.15–1.5 μg $mL^{-1}$), which suggests a role of moderate amounts of GAs in growth response to bacterial inoculation (Fig. 5). In contrast to the effect of BR 10268 broth, culture supernatant alone (in both cases after 100 times dilution in water) did not have

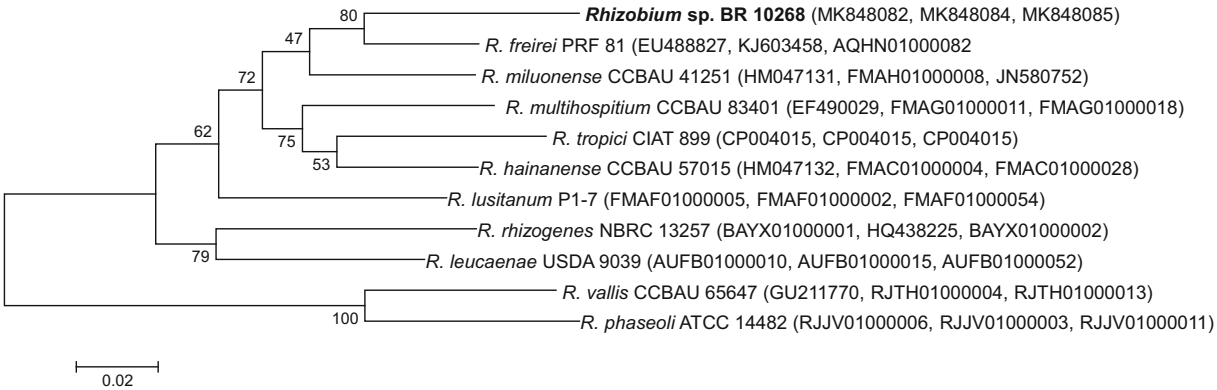


**Fig. 6** Molecular phylogenetic relationship of *Rhizobium* sp. BR 10268 with related *Rhizobium* type strains based on concatenated partial nucleotide sequence of housekeeping genes *recA* (386 bp), *gyrB* (577 bp) and *rpoB* (947 bp). Trees were constructed using Mega 6.0, using maximum likelihood method based on the general time reversible method, with 500 replicates. Percentages at which branch topologies were observed are indicated by numbers and nucleotide dissimilarity levels are indicated by a scale bar

any effect on mini-sett development (Fig. 5). This could be expected because, after dilution, hormone concentrations in the supernatant were probably below physiologically relevant concentrations. On the other hand, we showed that BR 10268 effectively colonized the surface and internal tissues of mini-setts, so apparently the presence of the bacteria was determinant to the observed growth promotion. Preparation of mini-setts involves cutting out single nodes, a process which opens up opportunities for bacteria to colonize internal stem tissues via the cutting edges. It seems likely that effective colonization permitted continuous production of GA and possibly other growth-modulating substances by BR 10268 after inoculation. The lowest $GA_3$ concentration tested to compare to the effect of inoculation was 0.15 μg $mL^{-1}$, which is approximately 15 times more than concentrations detected in defined culture supernatant. However, currently we do not know how this in vitro concentration relates to concentrations/quantities of GA effectively produced in a dynamic system such as the mini-sett tissue.

A natural important role of GAs on sugarcane shoot elongation and sprouting was reported decades ago, for example by Bull (1964) and reviewed by Vlitos (1974). Application of GA on sugarcane stubble in the field was found to improve sprouting (Kanwar and Kaur 1977). GAs were suggested to have the most pronounced effect on sugarcane growth during unfavorable conditions like drought and cold (Wittwer and Bukovac 1958). A recent study found beneficial effects of leave application of $GA_3$ (150 μg $mL^{-1}$) on yield and sucrose content in field-grown sugarcane (Nguyen et al. 2019). However, we are unaware of previous studies describing biometric details of sugarcane sprouting in response to varying gibberellin concentrations. The concentration of GAs detected in culture medium in the present study was higher than previously reported for *Rhizobium leguminosarum* bv. *phaseoli*, where the highest concentration detected was approximately 0.6 ng $GA_1$ $mL^{-1}$ (Atzorn et al. 1988). However, studies for *Herbaspirillum seropedicae* and *Gluconacetobacter diazotrophicus* have reported similar concentrations as the present study (Bastián et al. 1998). Another study even reported concentrations in the order of 100 ng $mL^{-1}$ for *Azospirillum* and *Arthrobacter* genera (Cacciari et al. 1989). In that study it was also shown that mixed cultures produce higher levels of GA and cytokinins than each bacterium cultivated individually, demonstrating that production of phytohormones can be increased in a more natural and complex environment, which may also be the case during plant-bacteria interaction. Therefore it is likely that quantities of phytohormones produced in situ in natural situations will differ from concentrations measured in defined culture media and possibly levels would be much higher in those conditions.

Apart from GAs, IAA and small amounts of the cytokinin trans-zeatin were also produced in defined medium. IAA has been associated with stimulation of lateral root initiation, and stem and root elongation, and is frequently involved as a mechanism of plant growth promotion by associative bacteria like *Azospirillum* spp. (Cassán et al. 2014). Cytokinins induce cell devisions and are involved in several developmental processes in both shoot and root tissues (Sakakibara 2006). Since at

least three classes of phytohormones are produced by BR 10268, it is not possible to attribute growth promotion effects to any of these classes. It is thus likely that these effects are a consequence of the interplay of the different growth modulating substances detected in this study and possibly additional factors. This complexity could also explain why shoot growth promotion during the sprouting phase was followed by root growth promotion later on during development (Table 1 and Figs. 2 and 3).

MLSA analysis of housekeeping genes *recA*, *gyrB* and *rpoB* confirmed that strain BR 10268 is related to *R. miluonense*, a species originally isolated from *Lespedeza* root nodules in China (Gu et al. 2008; Rouws et al. 2014). However, the more elaborate analysis performed here showed that it is more closely related to *R. freirei*, a common bean microsymbiont (Dall'agnol et al. 2013). Although a PCR product of the expected size (approximately 800 bp) could be obtained and cloned, sequencing showed that this amplicon was not *nifH*. Our previous report, based on PCR amplification, reported strain BR 10268 positive for *nifH* (Rouws et al. 2014). Based on the results from the present study, probably the strain does not contain this gene.

The use of synthetic consortia of PGPR strains with complementary beneficial traits, like BNF, phosphate solubilization, plant hormone production, among others, is a solution to the development of more efficient microbial inoculants adapted to diverse conditions found in the field. The present study indicates that rhizobia can colonize sugarcane plants and modulate their growth. *Rhizobium* sp. BR 10268 colonized sprouting mini-setts and had positive effects on shoot emission and root and shoot development in later stages of plant development, probably via phytohormone production and our results suggest that GAs play a role in this. Therefore, rhizobia, like strain BR 10268, should be considered as candidate microorganisms in future strategies to develop microbial inoculants to improve crop productivity in a sustainable manner.

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